

Phan Anh Nguyen, MSc in Data Science

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EDUCATION

University of Northumbria, United Kingdom

Sep 2022 – Sep 2023

MSc. Big Data and Data Science Technology

Machine Learning, Statistical Analysis, Business Analytics, Visualization, Deep Learning

University of Huddersfield, United Kingdom

Sep 2017 – Jun 2020

BSc. in Computer Science

AI, Computational Math, Algorithm, Parallel and Distributed Computing, Big Data

Chu Van An High School for the Gifted, Hanoi

Sep 2013 – Jun 2017

Class: Chemistry

PROFESSIONAL EXPERIENCE

FPT Smart Cloud (FCI)

Data Analyst, AI Engineer

Jan 2025 - Present

- Designed and implemented data-driven monitoring solutions for AI Agent performance through interactive dashboards, tracking key ML/LLM metrics such as token usage, message volume, user engagement, conversation topics, and sentiment/emotion classification.
- Partnered with the product and engineering teams to define scalable system architectures for LLM-based applications.
- Performed statistical analysis and data profiling to curate high-quality training datasets for both pre-training and fine-tuning phases, enabling improved model alignment with real-world conversational behavior and business objectives.

ELCOM Technology Communications Corporation

Data Analyst, Business Intelligence

Sep 2020 – June 2022

1SK- Mobile Gym Shop App Project:

- Developed automated data pipelines and SQL-based data models to support exploratory analysis, feature engineering, and business reporting on sales, customer retention, and gross merchandise value (GMV).
- Built interactive Power BI dashboards to visualize key performance indicators (KPIs), uncover marketing and sales trends, and enable data-driven decision-making for business stakeholders.
- Performed ad-hoc data analysis and delivered executive-ready reports to provide actionable insights into monthly performance, customer behavior, and campaign effectiveness, supporting strategic planning and operational improvements.

Traffic System (ITS) Project:

- Collaborated cross-functionally with AI and engineering teams to optimize machine learning models for real-time traffic violation detection, reducing inference latency and improving system responsiveness for smart transportation monitoring.

- Integrated legal compliance constraints into model features and rule-based systems, working closely with domain experts to ensure regulatory alignment—earning recognition from local authorities for accuracy and compliance.
- Led data-driven communication between stakeholders (BOT Son Hai) and the IT team to align technical development with business requirements for a smart traffic solution.
- Authored technical documentation and delivered stakeholder training to support smooth adoption and operational readiness during project handover and acceptance phases.

DATA SCIENCE PROJECTS

Customer Segmentation

- Supervised learning: Utilized boosting tree-based classifiers to model customer behavior and personas, applying SHAP for model interpretability and decision tree visualizations to translate findings into strategic business actions.
- Unsupervised learning: Leveraged Principal Component Analysis (PCA) for dimensionality reduction and K-means clustering to segment customers into distinct behavioral profiles.
- Statistical validation: Confirmed dataset suitability through Factor Analysis with Bartlett's test ($p < 0.001$) and KMO measure (>0.7).

Credit Default Classification

- Ensemble modeling: Developed a LightGBM classifier (AUC: 0.75, accuracy: 76%) to predict default probability, incorporating feature engineering for improved risk signals.
- Threshold tuning for profit: Calibrated decision threshold to 0.197 using cost-sensitive learning, maximizing net profit to £2.23M while balancing risk exposure.
- False negative minimization: Reduced Type II errors by 37% through class reweighting, directly cutting potential losses by £1.1M annually.

SKILLS & ADDITIONS

- Modelling: Sklearn, LightGBM, XGBoosting, Scipy, Tensorflow, Pytorch, Time Series Library.
- Data Analysis: Pandas, Numpy, Spark, Auto EDA tool (SweetViz,...)
- Data Visualization: Tableau, Power BI (DAX), Seaborn, Plotly.
- Database: PostgreSQL, Oracle SQL, BigQuery, Microsoft Azure, AWS.
- Soft skills: Proficient in communication, collaborative teamwork, stakeholder negotiation, and strategic planning.

Mastering Data Analytics Course (Hanoi)

Sep 2023 – Jan 2024

- Employed descriptive and diagnostic analytics to assess business performance.
- Developed professional dashboards in Power BI using a systematic approach.
- Performed EDA with dashboards and supporting analyses.

Ngo Vinh Coaching 1-on-1 Data Analyst Course (Hanoi)

Mar 2024 – Jun 2024

- Supervised and unsupervised ML via a credit approval prediction project.
- Core statistics and applied A/B testing techniques in real-world scenarios.

HONORS & AWARDS

- University of Northumbria scholarship for potential international students.
- University of Huddersfield scholarship Recipient with Academic Excellence and Potential.